

Looped boat propellers could help quiet the ocean for whales

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Whales use sound to find food and **communicate**. In quiet water, they can hear one another's calls from hundreds of miles away. But the ocean also carries noise from ships and other human activities.

Some shipping noise overlaps with the low sounds whales use. That can make important calls harder to hear.

One source of boat noise is the **propeller**, the spinning part that pushes a boat through water. Fast-moving **propellers** can create tiny bubbles that collapse and release sharp bursts of sound. This process is called **cavitation**.

A Detroit company, Sharrow Marine, has designed a different **propeller**. Instead of blades with open tips, its blades form twisting loops. The company says this reduces swirling water near the tips and creates less **cavitation**.

Sharrow reports that independent tests found reductions of up to seven **decibels** in underwater noise within the sound range whales use to **communicate**. **Decibels** measure sound levels. The amount of noise reduction can vary with the sound's **pitch** and how the equipment is used.

The company is now working with Ocean Alliance to test the **propellers** on research trips near whales. It is still early, so quieter measurements do not yet prove better lives for the animals.

The idea began with a different puzzle. Founder Greg Sharrow wanted to film a violinist using a **drone** without its loud buzzing. Thinking about quieter blades led him from flying machines to boats and a possible way to make the ocean easier to hear.

Word Treasure

communicate -- Share information with others.

propeller -- A spinning part that pushes a boat through water.

cavitation -- The forming and collapsing of tiny bubbles in water, which can make noise.

decibels -- Units used to measure sound levels.

pitch -- How high or low a sound is.

drone -- A flying machine without a pilot riding inside.

Background you need

A propeller spins to push a boat through water.

Decibels measure sound levels; pitch describes how high or low a sound is.

Break it down

Explains how whales use sound and how shipping noise can interfere.

Describes cavitation and the company's looped propeller design.

Presents the company's noise claims and the early tests near whales.

Ends with the drone-recording problem that inspired the idea.

Why it matters

Whales rely on sound. Testing quieter propellers could help researchers learn whether changing boat equipment reduces interference with whale communication.

Test what you remember

1. What do whales use sound to do?

- ☐ (A) Warm water and make bubbles
- ☐ (B) Move boats and guide drones
- ☐ (C) Find food and communicate
- ☐ (D) Change color and stay hidden

2. How can shipping noise affect whale calls?

- ☐ (A) It can make calls harder to hear.
- ☐ (B) It can make calls last longer.
- ☐ (C) It can make calls higher in pitch.
- ☐ (D) It can make calls travel farther.

3. What creates bursts of sound during cavitation?

- ☐ (A) Tiny bubbles collapsing
- ☐ (B) Drone cameras recording
- ☐ (C) Whale calls overlapping
- ☐ (D) Boat passengers speaking

4. What shape do Sharrow Marine's propeller blades form?

- ☐ (A) Flat squares
- ☐ (B) Straight rods
- ☐ (C) Twisting loops
- ☐ (D) Round disks

5. Who is working with Sharrow Marine to test the propellers near whales?

- ☐ (A) Greg's orchestra
- ☐ (B) Ocean Alliance
- ☐ (C) Boat passengers
- ☐ (D) Drone filmmakers

6. What first led Greg Sharrow to think about quieter propellers?

- ☐ (A) Hearing whale calls from a ship
- ☐ (B) Measuring noise on research trips
- ☐ (C) Filming a violinist with a drone
- ☐ (D) Testing bubbles in ocean water

Answer key (try the quiz first!): 1: C 2: A 3: A 4: C 5: B 6: C

MY THOUGHTS (at least 20 words):